SQL: Exercise 3

① SELECT product_id, product_name, price
CASE
WHEN price > 1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
WHEN price < 100 THEN 'Budget'
END AS price_tier
FROM products;

- THE PRODUCT TABLE

| Product_name | price | price_category |
|---|---|---|
| Laptop | 1200.00 | Expensive |
| Phone | 800.00 | Mid-range |
| Keyboard | 45.00 | Budget |
| Monitor | 300.00 | Mid-range |
| Mouse | 25.00 | Budge |

② SELECT
customer_name, amount
CASE
WHEN amount >= 1000 THEN 'High Value'
WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
WHEN amount < 500 THEN 'low Value'
END AS order_value_category
FROM orders;

| customer_name | amount | order_value_category |
|---|---|---|
| Alice | 150.00 | Low-Value |
| Bob | 560.00 | Medium value |
| Charlie | 999.99 | Medium value |
| Diana | 45.50 | Low Value |
| Ethan | 1200.00 | High Value |

③ SELECT emp_name, department, salary

```
CASE
    WHEN department = 'IT' AND Salary > 80 000
    THEN 'Senior IT'
WHEN department = 'HR' AND Salary > 55 000
THEN 'Experienced HR'
ELSE 'STAFF'
END AS position_level
FROM employees
```

employees Table

| emp_name | department | Salary | position |
|---|---|---|---|
| John | IT | 85 000 | Senior IT |
| Sara | HR | 60 00 | Experienced HR |
| Mark | IT | 75 000 | Staff |
| Lucy | Finance | 95 000 | Staff |
| Tom | HR | 55 000 | Staff |

④ SELECT student, name, score

```
    CASE
        WHEN score >= 90 THEN 'A'
        WHEN scae >= 80  THEN 'B'
        WHEN SCORE >= TO  THEN 'C'
        WHEN score >= 60  THEN 'D'
    ELSE 'F'
END AS grade
FROM Student;
```

Table: Student

| student_name | score | grade |
|---|---|---|
| Anna | 92 | A |
| Ben | 76 | C |
| Cara | 59 | F |
| David | 83 | B |
| Ella | 68 | D |

⑤
```
SELECT delievery_id, delievery_time_minutes,
    CASE
        WHEN delievery_time_minutes <= 30 THEN 'Fast'
        WHEN delievery_time_minutes <= 60 THEN 'onTime'
        ELSE 'Late'
END AS performance
FROM delieveries;
```

| delievery_id | delievery_time_minute | performance |
|---|---|---|
| 1 | 45 | On Time |
| 2 | 80 | Late |
| 3 | 30 | Fast |
| 4 | 65 | Late |
| 5 | 100 | Late |

⑥ SELECT

issue_type, priority

CASE

WHEN priority = 3 THEN 'High'

WHEN priority = 2 THEN 'Medium'

WHEN priority = 1 THEN 'Low'

END AS priority_label

FROM tickets

| issue_type | priority | priority_label |
|---|---|---|
| login_issue | 1 | low |
| Server down | 3 | High |
| Slow system | 2 | Medium |
| Email Error | 2 | Medium |
| Password reset | 1 | low |

⑦ SELECT student_id, days_present, total_days,

(days_present * 100.0 / total_days) AS attendance_percentage

CASE

WHEN (days_present * 100.0 / total_days) >= 90 THEN 'Excellent'

WHEN (days_present * 100.0 / total_days) >= 75 THEN 'Good'

ELSE 'Needs Improvement'

END AS attendance_Status

FROM attendance;

| student_id | attendance_percentage | attendance_Status |
|---|---|---|
| 1 | 90.00% | Excellent |
| 2 | 60.00% | Needs Improvement |
| 3 | 96.00% | Excellent |
| 4 | 50.00% | Needs Improvement |
| 5 | 100.00% | Excellent |

⑧
```
SELECT product_id, stock_qty,
CASE
    WHEN Stock_qty = 0 THEN 'out OF Stock'
    WHEN Stock_qty BETWEEN 1 AND 5 THEN 'LowStock'
    WHEN Stock_qty > 5 THEN 'In Stock'
END AS Stock_Status
FROM Products;
```

| Product_id | Stock_qty | Stock_status |
|---|---|---|
| 1 | 5 | Low Stock |
| 2 | 0 | Out of Stock |
| 3 | 25 | In Stock |
| 4 | 10 | In Stock |
| 5 | 3 | Low Stock |

⑨
```
SELECT class_id, subject, enrolled_students,
  CASE
    WHEN enrolled_student >= 25 THEN 'large'
    WHEN enrolled_student >= 10 THEN 'Medium'
    ELSE 'Small'
END AS Class_size_category
```

| subject | enrolled_student | class_size_category |
|---|---|---|
| Math | 30 | Large |
| English | 25 | Large |
| Science | 15 | Medium |
| Art | 5 | Small |
| History | 20 | Medium |

⑯
```
SELECT payment_id, payment_method, amount,
  CASE
      WHEN payment_method = 'cash' AND Amount >= 200
     THEN 'Eligible for Discount'
  ELSE 'Not Eligible'
 END AS discount_Eligibility
FROM payments;
```

| Payment_id | payment_method | amount | discount_eligibility |
|---|---|---|---|
| 1 | Card | 50 | Not Eligible |
| 2 | Cash | 200 | Eligible for discount |
| 3 | Card | 150 | Not Eligible |
| 4 | Paypal | 75 | Not Eligible |
| 5 | Cash | 300 | Eligible for Discount |